

CIVICPULSE AI

Phase-Wise Development Plan

Complete build tracker mapping every section and appendix of the 61-page
Community-Hero-Master-Plan.pdf to 20 sequential phases.

- Companion doc: Community-Hero-Master-Plan.pdf
- Product: Community Hero - Hyperlocal Problem Solver (Vibe2Ship)
- Deadline: June 29, 2026, 2:00 PM (BlockseBlock submission)
- Phases: 0 through 19 (20 phases total)
- Usage: Complete each phase fully before starting the next.
- Track progress: Check [] boxes as you build with Cursor + AI Studio.

Generated: June 23, 2026

MASTER PLAN COVERAGE INDEX (100% MAPPING)

Every section (1-36) and appendix (A-W) from Community-Hero-Master-Plan.pdf is assigned to exactly one primary phase below. No content is omitted.

ID	Master Plan Content	Phase
Section 1	Executive Summary	Phase 0
Section 2	Hackathon Problem Statement (all 2.1-2.4 + 8 features)	Phase 0, spread across build phases
Section 3	Competitive & Research Landscape	Phase 0
Section 4	Recommended Solution Architecture	Phase 1
Section 5.1	Image & Video Reporting	Phase 2
Section 5.2	AI Issue Categorization	Phase 2
Section 5.3	Geo-location & Mapping	Phase 3
Section 5.4	Community Verification	Phase 5
Section 5.5	Real-time Issue Tracking	Phase 4, 7
Section 5.6	Impact Dashboards	Phase 8
Section 5.7	Predictive Insights	Phase 9
Section 5.8	Gamification	Phase 10
Section 6	Agentic AI Design (6 agents + orchestration + tools)	Phase 6, 11
Section 7	Google Technologies Stack	Phase 1, spread all phases
Section 8	Google AI Studio Deployment Strategy	Phase 1, 16, 18
Section 9	MCP Toolchain Integration	Phase 1, 15, 16
Section 10	Data Model (Firestore)	Phase 4
Section 11	Open311 Interoperability	Phase 12
Section 12	Evaluation Matrix Alignment	Phase 0, 18
Section 13	7-Day Execution Roadmap	Mapped across Phases 1-19
Section 14	Submission Deliverables Checklist	Phase 18
Section 15	AI Studio Starter Prompts (6 prompts)	Phases 2-11
Section 16	Risks & Mitigations	Phase 13, 16
Section 17	Reference Repositories & Links	Phase 0, 15
Section 18	Conclusion	Phase 19
Section 19	LogiFlow Benchmark Quality Bar	Phase 0, 15
Section 20	System Design Principles	Phase 1, all build phases
Section 21	Full System Architecture	Phase 1, 15

ID	Master Plan Content	Phase
Section 22	Frontend Architecture (18 routes)	Phases 3-14
Section 23	9 Civic Pipelines	Phases 2-12
Section 24	Report Submission Lifecycle (10 steps)	Phases 2, 6
Section 25	Performance & Caching L1-L5	Phase 13
Section 26	Security & Rate Limiting	Phase 13
Section 27	Full API Contract (20+ endpoints)	Phases 2-12, 15
Section 28	Database Schema + Indexes	Phase 4
Section 29	ML & AI Models (A-G)	Phases 2, 5, 6, 9, 11
Section 30	Repository Structure	Phase 1, 15
Section 31	Deployment Runbook	Phase 16, 18
Section 32	16 Diagram Specifications	Phase 15
Section 33	Testing & QA Strategy	Phase 17
Section 34	Business Impact & USP	Phase 0, 18
Section 35	Makefile & Automation	Phase 16
Section 36	Hour-by-Hour 7-Day Sprint	Phases 1-19 timeline
Appendix A	Evaluation Matrix (official weights)	Phase 0, 18
Appendix B	Gemini Model Selection Guide	Phases 2, 6, 9, 11
Appendix C	REST API Specification (10 endpoints)	Phases 2-12
Appendix D	Gemini Structured Output Schema	Phase 2
Appendix E	Screen-by-Screen UX (11 screens)	Phases 3-14
Appendix F	Competitive Feature Matrix	Phase 0, 18
Appendix G	Firestore Security Rules	Phase 4, 13
Appendix H	AI Pipeline Server Pseudocode	Phase 2, 6
Appendix I	3-Minute Judge Demo Script	Phase 19
Appendix J	Google Doc Submission Template	Phase 18
Appendix K	POST /api/reports Request/Response Schema	Phase 2
Appendix L	SLA Matrix by Category & Severity	Phase 7
Appendix M	Open311 Service Code Mapping	Phase 12
Appendix N	Environment Variables (complete)	Phase 1, 16
Appendix O	Gamification Point Economy	Phase 10
Appendix P	Presentation Kit (15 slides)	Phase 18
Appendix Q	LogiFlow vs CIVICPULSE AI Comparison	Phase 0, 15
Appendix R	Seed Data Specification (25 issues)	Phase 17
Appendix S	Mermaid 01-System-Architecture Source	Phase 15
Appendix T	TODO.md Master Checklist	All phases (rolling)
Appendix U	Team Role Assignments	Phase 0, 1

ID	Master Plan Content	Phase
Appendix V	Citizen Assistant Function Calling Tools	Phase 11
Appendix W	Error Codes & HTTP Status	Phase 13

PHASE TIMELINE vs 7-DAY SPRINT (SECTION 36)

Phase	Title	Sprint Day	Hours
0	Foundation	Day 0	2-3h
1	Scaffold & GCP	Day 1 AM	4h
2	Report + Vision AI	Day 1 PM - Day 2 AM	10h
3	Maps & Geo	Day 2 PM - Day 3 AM	8h
4	Firestore Realtime	Day 3 PM - Day 4 AM	6h
5	Community Verify	Day 4 AM	6h
6	6 Agents	Day 4 PM	6h
7	Admin & SLA	Day 4 PM - Day 5 AM	4h
8	Dashboards	Day 5 AM	6h
9	Predictive Insights	Day 5 PM - Day 6 AM	4h
10	Gamification	Day 5 PM	3h
11	AI Chat	Day 6 AM	3h
12	Threads & Open311	Day 6 AM	3h
13	Security & Perf	Day 6 AM-PM	4h
14	Frontend Polish	Day 6 PM	4h

Phase	Title	Sprint Day	Hours
15	Docs & Diagrams	Day 6 PM	4h
16	Deploy & CI/CD	Day 6 PM - Day 7 AM	4h
17	Testing & Seed	Day 7 AM	3h
18	Submission	Day 7 AM-2PM	3h
19	Demo & Closure	Day 7 + eval period	2h+

OFFICIAL 8 HACKATHON FEATURES -> PHASE MAP

#	Official Feature (Section 2.3)	Primary Phase	Also Covered
1	Image and video-based reporting	Phase 2	Phase 14 UI
2	AI-powered issue categorization	Phase 2	Phase 6 Agent 2
3	Geo-location and mapping	Phase 3	Phase 4 geo fields
4	Community verification	Phase 5	Phase 10 points
5	Real-time issue tracking	Phase 4, 7	Phase 6 notify
6	Impact dashboards	Phase 8	Phase 9 insights
7	Predictive insights	Phase 9	Phase 6 Agent 6
8	Gamification	Phase 10	Phase 14 leaderboard UI

HOW TO USE THIS DOCUMENT

1. Complete phases in order 0 -> 19. Do not skip.
2. For each phase: read listed Master Plan sections in Community-Hero-Master-Plan.pdf.
3. Execute every Build Task checkbox.
4. Verify all Definition of Done items before advancing.
5. Run Verification checklist - if any fail, stay in current phase.
6. Mark phase complete in Rolling Master Checklist at end of this PDF.
7. Use Cursor + AI Studio + MCP tools referenced in Master Plan Section 9.

PHASE 0: FOUNDATION, REQUIREMENTS & QUALITY BAR

Goal: Lock scope, evaluation criteria, competitive positioning, and LogiFlow-grade quality expectations before writing code.

Suggested timing: Day 0 (before build) | ~2-3 hours

Depends on: None - read Master Plan Sections 1-3, 12, 18, 19, 34 and Appendices A, F, Q

Master Plan Coverage (sections + appendices)

- Master Plan Section 1 - Executive Summary (CIVICPULSE AI codename, strategic thesis)
- Master Plan Section 2.1-2.4 - Background, Challenge, 8 Example Features, Evaluation Focus
- Master Plan Section 3 - FixMyStreet, InfraGuard, Swachhata, CivicThreads, all research refs
- Master Plan Section 12 - Evaluation Matrix Alignment (20% agentic, 20% impact, etc.)
- Master Plan Section 18 - Conclusion and winning formula
- Master Plan Section 19 - LogiFlow benchmark (771 commits, 6275 doc lines, 16 diagrams)
- Master Plan Section 34 - Business Impact & USP (80% faster triage, 40% fewer duplicates)
- Appendix A - Official evaluation weights table
- Appendix F - Competitive matrix vs Swachhata, FixMyStreet, InfraGuard
- Appendix Q - LogiFlow vs CIVICPULSE AI comparison
- Appendix U - Team role assignments (Ojas = Tech Lead)

Official Hackathon Features Addressed This Phase

- All 8 features listed as MVP scope - assign each to build phase mentally

Build Tasks (complete every item)

- ☐ Read entire Community-Hero-Master-Plan.pdf cover to cover (61 pages)
- ☐ Confirm problem statement: Community Hero - Hyperlocal Problem Solver
- ☐ Write 1-paragraph product vision mirroring Section 1 executive summary
- ☐ List all 8 official example features and note which phase builds each
- ☐ Memorize evaluation weights: Impact 20%, Agentic 20%, Innovation 20%, Google Tech 15%
- ☐ Identify differentiation vs Swachhata: agentic AI, predictive hotspots, sub-3s vision triage
- ☐ Assign team roles per Appendix U (Tech Lead, AI Engineer, Frontend, Data/Geo, DevOps)
- ☐ Create shared tracking doc or use this PDF checkboxes per phase
- ☐ Bookmark reference repos from Section 17 (InfraGuard, PotSoft, CivicThreads, etc.)
- ☐ Set calendar deadline: June 29, 2026 2:00 PM BlockseBlock final submit

Deliverables

- Team aligned on scope, name (CIVICPULSE AI), and quality bar
- Phase tracker started (this PDF with checkboxes)
- Roles assigned

Definition of Done

- ☐ Every team member read Master Plan Sections 1-3 and 12
- ☐ All 8 hackathon features mapped to phases 2-12
- ☐ Evaluation criteria understood and design decisions will target Agentic Depth 20%

Verification Before Next Phase

- ☐ Can explain product in 30 seconds (photo -> AI -> map -> verify -> resolve)
- ☐ Can name 3 competitor apps and 3 differentiators

☐ Deadline and submission requirements (Section 14) understood

PHASE 1: PROJECT SCAFFOLD & GOOGLE CLOUD FOUNDATION

Goal: Create AI Studio app, Firebase backend, GitHub repo, docs skeleton, and dev environment matching Section 30 repo structure.

Suggested timing: Day 1 hours 0-4 | Master Plan Section 36 Day 1 H0-H4

Depends on: Phase 0 complete

Master Plan Coverage (sections + appendices)

- Section 4 - Recommended Solution Architecture (4.1-4.4 latency budget)
- Section 7 - Google Technologies Stack (core: AI Studio, Gemini, Cloud Run, Firestore, Auth, Storage)
- Section 8.1 - AI Studio environment setup (Build mode, React + server runtime)
- Section 9 - MCP toolchain (firebase_login, gcloud, github, browser)
- Section 20 - System design principles (modular pipelines, separation of concerns)
- Section 21 - Production topology overview
- Section 30 - Repository structure (docs/, server/, scripts/, Makefile)
- Section 35 - Makefile targets (dev, seed, deploy, diagrams)
- Appendix N - All environment variables documented in .env.example
- Appendix T - Copy TODO.md master checklist into repo root

Build Tasks (complete every item)

- [] Open aistudio.google.com -> Build mode -> New app -> React + Node server runtime
- [] System instruction: 'Build Community Hero civic reporting platform per master plan'
- [] Prompt: 'Add Firestore database and Firebase Auth with Google Sign-In' (Section 8.1 step 3)
- [] Add Secrets panel: GEMINI_API_KEY (auto), MAPS_API_KEY if needed (Appendix N)
- [] Export to GitHub early - create repo community-hero or civicpulse-ai
- [] Clone to /Users/ojas/Desktop/Vibe2Ship and open in Cursor
- [] Scaffold docs/ folder: architecture.md, system-design.md, deployment.md, api_contract.md (stubs)
- [] Create firestore.rules, firestore.indexes.json, storage.rules (starter from Appendix G)
- [] Create .env.example with all Appendix N variables
- [] Create Makefile with targets: dev, seed, deploy, diagrams, lint, test, health (Section 35)
- [] Run firebase_login via MCP if using Firebase MCP tools
- [] Verify gcloud auth: project-6d6f652b-7066-4341-806, region asia-south1
- [] Create README.md with problem statement, stack, and links placeholders
- [] Initialize TODO.md from Appendix T (all items unchecked)

Deliverables

- AI Studio app with Firestore + Auth enabled
- GitHub repository with docs/ skeleton and Makefile
- Local Cursor workspace synced
- .env.example and firestore.rules committed

Definition of Done

- [] AI Studio Build mode loads without errors
- [] Google Sign-In works in dev container
- [] GitHub repo exists and is pushed
- [] docs/ and scripts/ folders match Section 30 structure

Verification Before Next Phase

- ☐ Firebase Auth login/logout works
- ☐ Firestore write test document succeeds
- ☐ README lists Google stack from Section 7

PHASE 2: REPORT INTAKE + GEMINI VISION AI PIPELINE

Goal: Ship core report flow: camera, GPS, image upload, Gemini 2.5 Flash structured classification, POST /api/reports.

Suggested timing: Day 1 H4-12 + Day 2 H0-6 | Section 36 Day 1-2

Depends on: Phase 1 complete

Master Plan Coverage (sections + appendices)

- Section 5.1 - Image & video reporting (camera, gallery, compress 1280px WebP)
- Section 5.2 - AI categorization (Gemini JSON schema, 9 categories, confidence)
- Section 15 Prompt 1 - Scaffold PWA report wizard
- Section 15 Prompt 2 - Gemini 2.5 Flash server-side analysis
- Section 23 Pipelines 1-2 - Intake + Vision pipelines
- Section 24 Steps 1-5 - Client capture through VisionAgent
- Section 27 - POST /api/reports, POST /ai/analyze endpoints
- Section 29 Model A - Gemini 2.5 Flash vision classifier
- Appendix B - Use gemini-2.5-flash NOT flash-image for JSON
- Appendix C - POST /api/reports in API spec
- Appendix D - Full structured output schema (category, severity, department, etc.)
- Appendix H - createReport pseudocode implementation
- Appendix K - Request/response multipart schema
- Appendix E Screens 2-4 - Report wizard steps

Official Hackathon Features Addressed This Phase

- Feature 1: Image and video-based issue reporting (photo path; video in Phase 2 stretch)
- Feature 2: AI-powered issue categorization

Build Tasks (complete every item)

- [] Build 3-step Report Wizard UI (Appendix E screens 2-4)
- [] Implement camera capture + gallery upload + HTML5 Geolocation
- [] Client-side image resize max 1280px WebP before upload (Section 4.4)
- [] Implement POST /api/reports multipart handler (Appendix K)
- [] Upload images to Cloud Storage path issues/{id}/original.jpg (Section 28)
- [] Implement VisionAgent with gemini-2.5-flash + responseType application/json
- [] Define Zod/IssueAnalysisSchema matching Appendix D all fields
- [] Pre-fill report form from AI output; user can edit before submit
- [] Confidence gate: if confidence < 0.6 return needs_review status (Section 20.3)
- [] Tag aiMetadata.data_source = 'ai' on all AI fields
- [] Implement GET /api/health endpoint
- [] Optional: 15s video upload + 3 keyframe extraction (Section 5.1 video)
- [] Write docs/pipelines/intake.md and docs/pipelines/vision.md
- [] Latency target: Gemini classify < 2s (Section 4.3 budget)

Deliverables

- Working report wizard with photo + GPS
- Gemini returns structured JSON with category, severity, department
- Issue saved to Firestore issues collection
- api_contract.md documents POST /api/reports

Definition of Done

- ☐ End-to-end: photo -> AI analysis -> editable form -> Firestore write
- ☐ P95 classify latency < 4s on test image
- ☐ Invalid/blank images rejected with InvalidMediaCard (Section 20.4)

Verification Before Next Phase

- ☐ Test with pothole, waste, streetlight sample images
- ☐ JSON validates against Appendix D schema
- ☐ GEMINI_API_KEY never exposed in browser network tab

PHASE 3: GEO-LOCATION, MAPS & ISSUE DISCOVERY UI

Goal: Google Maps integration, marker clustering, geocoding, landing page, map explorer, issue list.

Suggested timing: Day 2 H6-12 + Day 3 | Section 36 Day 2-3

Depends on: Phase 2 complete (issues exist in Firestore)

Master Plan Coverage (sections + appendices)

- Section 5.3 - Geo-location & mapping (Maps JS API, MarkerClusterer, Geocoding)
- Section 15 Prompt 3 - Maps integration prompt
- Section 22 Routes: /, /map, partial /issues/:id
- Section 23 Pipeline 3 - Geo pipeline (reverse geocode, ward, geohash-7)
- Section 24 Step 3 - parallel reverse-geocode
- Section 27 - GET /api/reports geospatial query
- Appendix E Screens 1, 6 - Landing + Map Explorer
- Appendix N - MAPS_API_KEY restricted to Cloud Run URL

Official Hackathon Features Addressed This Phase

- Feature 3: Geo-location and mapping

Build Tasks (complete every item)

- [] Integrate Google Maps JavaScript API + Advanced Markers
- [] Add @googlemaps/markerclusterer with supercluster (Section 5.3)
- [] Color markers by severity: red/orange/yellow/green
- [] Reverse geocode lat/lng to address on report submit (Geo pipeline)
- [] Store GeoPoint + geohash + address on issue doc (Section 28)
- [] Build Landing page / with live issue count, map preview, Report CTA (Appendix E screen 1)
- [] Build Map Explorer /map with filters: category, status, severity
- [] Implement GET /api/reports?lat=&lng=&radius_km=2&status=open
- [] Click marker -> issue preview card with link to detail
- [] Places Autocomplete fallback when GPS weak (Section 5.3)
- [] Manual pin drop on map if GPS unavailable (Section 20.3)
- [] Create geocode_cache collection for TTL caching (Section 28)
- [] Write docs/pipelines/geo.md

Deliverables

- Map shows all submitted issues with clustering
- Landing page with stats and CTA
- Issues have human-readable addresses

Definition of Done

- [] Map loads on mobile Safari/Chrome over HTTPS
- [] Filters correctly hide/show issue categories
- [] Empty state: 'Be the first reporter' when no issues (Section 20.4)

Verification Before Next Phase

- [] 10 test issues visible on map with correct colors
- [] Geocode returns sensible address for test coordinates
- [] Maps API key restricted in Google Cloud Console

PHASE 4: FIRESTORE DATA LAYER, REALTIME SYNC & ISSUE DETAIL

Goal: Complete schema, security rules, realtime listeners, issue detail page, my reports, status timeline.

Suggested timing: Day 3 H6-12 + Day 4 morning | Section 36 Day 3-4

Depends on: Phase 2-3 complete

Master Plan Coverage (sections + appendices)

- Section 5.5 - Real-time tracking (onSnapshot, status machine, timeline)
- Section 10 - Data Model all collections
- Section 22 Routes: /issues/:id, /my-reports, /login, /profile
- Section 28 - Full database schema, composite indexes, storage paths
- Appendix G - Firestore security rules (full implementation)
- Appendix E Screens 5, 7 - Issue detail + My Reports
- Section 20.4 - Honest empty states on my reports

Official Hackathon Features Addressed This Phase

- Feature 5: Real-time issue tracking (partial - timeline + statuses)

Build Tasks (complete every item)

- [] Implement full Firestore schema from Section 28 (issues, users, departments, events subcollection)
- [] Deploy firestore.indexes.json composite indexes (status+createdAt, wardId+status, etc.)
- [] Implement firestore.rules from Appendix G with admin custom claims
- [] Status machine: Draft -> Submitted -> Verified -> Assigned -> In Progress -> Resolved -> Closed
- [] Subcollection issues/{id}/events for audit timeline
- [] Firestore onSnapshot on issues collection for realtime map updates
- [] Build Issue Detail page /issues/:id with photo, timeline, share link (Appendix E screen 5)
- [] WhatsApp-shareable public URL /issues/{id}
- [] Build My Reports /my-reports with status chips (Appendix E screen 7)
- [] Build Login /login and Profile /profile with useAuthStore (Section 22)
- [] Seed departments collection with SLA hours (prep for Phase 7 Appendix L)
- [] Implement GET /api/reports/:id with events array
- [] Create users collection on first login with civicPoints: 0

Deliverables

- Realtime map updates when new issue submitted
- Issue detail with full status timeline
- Firestore rules deployed and tested

Definition of Done

- [] Status transitions write events subcollection entries
- [] Unauthorized users cannot update others' issues
- [] My Reports shows only current user's issues

Verification Before Next Phase

- [] Open two browser tabs - new issue appears without refresh
- [] Security rules reject unauthenticated writes in Firebase emulator or prod test
- [] Share link opens issue detail for anonymous reader

PHASE 5: COMMUNITY VERIFICATION & DUPLICATE DETECTION

Goal: Upvote system, verification tiers, dedup agent with geohash + embeddings, merge UX.

Suggested timing: Day 4 H0-6 | Section 36 Day 4

Depends on: Phase 4 complete

Master Plan Coverage (sections + appendices)

- Section 5.4 - Community verification (upvote, tiers 1/3/10, anti-gaming)
- Section 15 Prompt 4 - Community upvote + dedup prompt
- Section 23 Pipelines 4-5 - Dedup + Verification pipelines
- Section 29 Model C - gemini-embedding-001 for duplicate detection
- Section 27 - POST /api/reports/:id/upvote, POST merge endpoint
- Appendix C - upvote endpoint

Official Hackathon Features Addressed This Phase

- Feature 4: Community verification

Build Tasks (complete every item)

- [] Implement POST /api/reports/:id/upvote idempotent (votes subcollection doc id = userId)
- [] Verification tiers: 1 upvote = Acknowledged, 3 = Community Verified, 10 = Priority Escalation
- [] Rate limit 30 upvotes/hour per user (Section 5.4)
- [] Require auth for upvote; one vote per user per issue
- [] Implement DedupAgent: 50m geohash query + embedding cosine > 0.85 (Section 5.4)
- [] Generate embedding via gemini-embedding-001 on report submit (Section 29 Model C)
- [] If duplicate: return duplicate_suggestion payload (Appendix K response)
- [] Build merge UX: POST /api/reports/:id/merge with target_issue_id
- [] Anti-gaming: account age or 1 prior report required to upvote (Section 26)
- [] Update verificationLevel field on issue doc when thresholds crossed
- [] Display verification badges on issue cards and detail page
- [] Write docs/pipelines/verification.md and dedup notes in vision.md

Deliverables

- Upvote button functional with live count
- Community Verified badge at 3 upvotes
- Duplicate suggestion when reporting same pothole twice nearby

Definition of Done

- [] Second account upvote increments count in realtime
- [] Duplicate report within 50m suggests merge rather than duplicate pin spam
- [] Rate limit returns 429 with Retry-After (Appendix W)

Verification Before Next Phase

- [] Demo: two devices upvote same issue -> badge appears
- [] Submit same photo at same GPS -> merge suggestion shown

PHASE 6: AGENTIC AI ORCHESTRATION (6 AGENTS)

Goal: Implement all 6 agents with deterministic workflow graph, routing, SLA, notifications.

Suggested timing: Day 4 H6-12 | Section 36 Day 4

Depends on: Phase 2, 5 complete

Master Plan Coverage (sections + appendices)

- Section 6 - Full agentic design (6.1 six agents, 6.2 orchestration, 6.3 tools partial)
- Section 23 Pipelines 4, 6, 9 - Routing, Notification pipelines
- Section 24 Steps 6-10 - Dedup through NotifyAgent
- Section 29 Models D-G - Priority score, SLA predictor formulas
- Appendix H - Full agent chain after createReport
- Appendix B - gemini-2.5-flash-lite for status narratives

Official Hackathon Features Addressed This Phase

- Evaluation focus: Agentic Depth 20% - primary scoring phase

Build Tasks (complete every item)

- [] Agent 1 Intake Triage: SafeSearch + is_civic_issue validation (Section 6.1)
- [] Agent 2 Vision Classifier: already in Phase 2 - wire into orchestrator
- [] Agent 3 Dedup & Cluster: Phase 5 - wire into orchestrator
- [] Agent 4 Routing & SLA: assign departmentId, compute slaDeadline from Appendix L matrix
- [] Agent 5 Citizen Communicator: generate plain-language status messages EN (+ HI stretch)
- [] Agent 6 Insights Analyst: stub for Phase 9 nightly batch
- [] Implement workflow graph: confidence < 0.7 -> review queue branch (Section 6.2)
- [] Priority score formula Section 29 Model G: $\text{severity} \times 0.4 + \text{upvotes} \times 0.2 + \text{safety} \times 0.3 + \text{age} \times 0.1$
- [] NotifyAgent: write notification on status change (in-app; FCM stretch)
- [] Implement runAgents() orchestrator called after every report create
- [] Log agent steps to issues/{id}/events with type ai_analysis, routing, notify
- [] Template-based status messages by default; Gemini lite for detailed mode (Section 25)
- [] Document agent workflow in docs/architecture.md matching Section 6

Deliverables

- 6 agents wired in server orchestrator
- Every new report gets department, SLA deadline, priority score automatically
- Agent execution logged in issue events timeline

Definition of Done

- [] Low confidence report goes to review queue not public map
- [] SLA deadline visible on issue detail
- [] Agentic workflow diagram ready for docs (content for Phase 15)

Verification Before Next Phase

- [] Trace one report through all 6 agents in events subcollection
- [] Department assignment matches category taxonomy Section 5.2

PHASE 7: ADMIN WORKFLOW, RESOLUTION & SLA ENFORCEMENT

Goal: Authority admin panel, status updates, proof photo, SLA breach tracking, reopen flow.

Suggested timing: Day 4 H9-12 + Day 5 morning | Section 36 Day 4-5

Depends on: Phase 6 complete

Master Plan Coverage (sections + appendices)

- Section 5.5 - Admin status updates, resolved proof photo, before/after AI optional
- Section 22 Routes: /admin, /admin/analytics (partial)
- Section 23 Pipeline 7 - Resolution pipeline
- Section 27 - POST /api/reports/:id/status, POST reopen
- Appendix L - Full SLA matrix by category and severity
- Appendix E Screen 10 - Admin panel
- Appendix W - FORBIDDEN 403 for non-admin

Official Hackathon Features Addressed This Phase

- Feature 5: Real-time issue tracking (complete with admin resolution loop)

Build Tasks (complete every item)

- [] Implement admin role via Firebase custom claims or ADMIN_UIDS env (Appendix N)
- [] Build Admin Queue /admin sorted by priorityScore desc
- [] Implement POST /api/reports/:id/status { status, note, proof_image }
- [] Upload proof photo to issues/{id}/proof.jpg
- [] Optional: Gemini before/after comparison Section 29 Model F
- [] SLA breach flag when now > slaDeadline and status not Resolved
- [] Dashboard tile for SLA breaches (prep Phase 8)
- [] Implement POST /api/reports/:id/reopen for unsatisfied citizens
- [] Status notifications via Agent 5 on each transition
- [] Bulk status update for demo (select multiple issues)
- [] Export CSV of open issues from admin panel
- [] Implement Appendix L SLA hours table in routing agent config
- [] Escalation: SLA breach -> priorityScore +25 (Appendix L note)

Deliverables

- Admin can mark In Progress -> Resolved with proof photo
- Citizen sees status timeline update in realtime
- SLA countdown on issue detail for open issues

Definition of Done

- [] Non-admin gets 403 on status update endpoint
- [] Resolved issue shows proof image on detail page
- [] Reopen flow returns issue to Submitted status

Verification Before Next Phase

- [] Full demo path: report -> verify -> admin resolve -> citizen sees Resolved
- [] SLA breach flagged on intentionally backdated test issue

PHASE 8: IMPACT DASHBOARDS & ANALYTICS

Goal: Citizen + admin dashboards, KPIs, charts, ward breakdown, cached aggregates.

Suggested timing: Day 5 H0-6 | Section 36 Day 5

Depends on: Phase 7 complete (resolved issues for metrics)

Master Plan Coverage (sections + appendices)

- Section 5.6 - Impact dashboards (KPIs, heatmaps, trends, SLA compliance)
- Section 15 Prompt 5 - Dashboard prompt
- Section 23 Pipeline 8 - Insights pipeline (aggregates portion)
- Section 27 - GET /api/analytics/summary, /hotspots, /trends
- Appendix E Screen 8 - Impact dashboard
- Section 25 - analytics_daily collection, 15min cache

Official Hackathon Features Addressed This Phase

- Feature 6: Impact dashboards

Build Tasks (complete every item)

- ☐ Build Impact Dashboard /dashboard with KPI tiles: open, resolved, avg resolution time
- ☐ Category pie chart (Chart.js or Recharts)
- ☐ 7-day and 30-day trend line charts
- ☐ Ward breakdown table if wardId populated
- ☐ Department SLA compliance percentage tile
- ☐ Citizen engagement metric: reports per day, upvotes per day
- ☐ Implement analytics_daily Firestore aggregation (Cloud Function or server cron)
- ☐ GET /api/analytics/summary with 15min cache (Section 25 L3)
- ☐ Build Admin Analytics /admin/analytics with heatmap view
- ☐ AI narrative card placeholder (filled Phase 9)
- ☐ Honest empty state when no data yet

Deliverables

- Dashboard loads real metrics from Firestore
- Charts update when new issue resolved
- Admin analytics shows category breakdown

Definition of Done

- ☐ KPI numbers match manual Firestore count verification
- ☐ Dashboard performs acceptably with 25+ seed issues

Verification Before Next Phase

- ☐ Resolve 3 issues -> resolved count increments on dashboard
- ☐ Export or screenshot dashboard for Google Doc submission

PHASE 9: PREDICTIVE INSIGHTS & HOTSPOT INTELLIGENCE

Goal: Hotspot detection, trend analysis, Gemini narrative insights, predictive cards.

Suggested timing: Day 5 H6-9 + Day 6 morning | Section 36 Day 5-6

Depends on: Phase 8 complete

Master Plan Coverage (sections + appendices)

- Section 5.7 - Predictive insights (hotspots, trends, recurring issues, preventive zones)
- Section 23 Pipeline 8 - Insights pipeline (full)
- Section 29 Models D-E - HotspotScorer v1, SLA Predictor v1
- Section 6.1 Agent 6 - Insights Analyst nightly batch
- Appendix R - Hotspot cells in seed data spec

Official Hackathon Features Addressed This Phase

- Feature 7: Predictive insights

Build Tasks (complete every item)

- [] Implement HotspotScorer v1: geohash-6 cells with >5 open issues in 7 days flagged (Section 5.7)
- [] Severity weight + recency decay in risk score (Section 29 Model D)
- [] Collection hotspots/{geohash} with issueCount, predictedRisk, updatedAt
- [] GET /api/analytics/hotspots?ward_id=
- [] Trend agent: weekly Gemini summary of emerging categories (gemini-2.5-flash-lite)
- [] GET /api/analytics/trends with narrative + structured data
- [] Dashboard AI insight card with Gemini-generated ward summary
- [] Recurring issue detection: same category same geohash-6 within 30 days
- [] Seasonal waste spike alert rule (demo-grade)
- [] Suggested preventive maintenance zones list on dashboard
- [] Agent 6 nightly cron aggregates and writes insights (Cloud Scheduler stretch)
- [] Document Models D-E in docs/system-design.md

Deliverables

- Hotspot list on dashboard with map overlay
- AI insight card with plain-language trend summary
- Predictive section visible to judges on demo

Definition of Done

- [] Cluster of 5+ issues triggers hotspot alert
- [] Gemini narrative mentions actual category counts from data not hallucinated

Verification Before Next Phase

- [] Seed 6 issues in small area -> hotspot appears
- [] Trend API returns text referencing real Firestore aggregates

PHASE 10: GAMIFICATION & CIVIC ENGAGEMENT

Goal: Points, badges, opt-in leaderboard, streaks, ethics guardrails.

Suggested timing: Day 5 H9-12 | Section 36 Day 5

Depends on: Phase 5, 7 complete

Master Plan Coverage (sections + appendices)

- Section 5.8 - Gamification (points, badges, leaderboard, ethics)
- Appendix O - Full point economy table
- Appendix E Screen 9 - Leaderboard
- Section 27 - GET /api/leaderboard

Official Hackathon Features Addressed This Phase

- Feature 8: Gamification for citizen engagement

Build Tasks (complete every item)

- ☐ Implement civicPoints on users collection (Appendix O values)
- ☐ +10 quality report conf>0.8, +15 Community Verified, +15 merge, +5 upvote, +25 resolved
- ☐ Badges array on user: First Report, Neighborhood Voice, Duplicate Hunter, etc.
- ☐ Award badges on trigger events server-side
- ☐ Build Leaderboard /leaderboard opt-in via leaderboardOptIn field
- ☐ GET /api/leaderboard?period=weekly|alltime
- ☐ 7-day reporting streak detection (+30 points)
- ☐ Profile page shows private badges and points by default (Section 5.8 ethics)
- ☐ Transparent rules page /terms or /gamification-rules
- ☐ No public trust score affecting issue credibility (Appendix O ethics note)
- ☐ Display points earned toast on report submit and upvote

Deliverables

- Users earn points and badges through real actions
- Opt-in leaderboard shows weekly champions
- Ethics rules page published

Definition of Done

- ☐ Gamification cannot be gamed without real reports (rate limits)
- ☐ Leaderboard hidden unless user opts in

Verification Before Next Phase

- ☐ Submit report -> +10 points on profile
- ☐ 3 upvotes on issue -> reporter gets +15 Neighborhood Voice badge

PHASE 11: CITIZEN AI ASSISTANT (FUNCTION CALLING CHAT)

Goal: Chat assistant with Gemini function calling - never hallucinates issue data.

Suggested timing: Day 6 H0-3 | Section 36 Day 6

Depends on: Phase 4, 9 complete

Master Plan Coverage (sections + appendices)

- Section 6.3 - Function calling tools list
- Section 15 Prompt 6 - Agent chat prompt
- Section 27 - POST /api/ai/chat
- Section 29 - Chat uses gemini-2.5-flash-lite with tools
- Appendix V - All 7 tool definitions
- Appendix E Screen 11 - AI Assistant Chat
- Appendix B - flash-lite for chat

Official Hackathon Features Addressed This Phase

- Evaluation: Agentic Depth - tool-grounded assistant

Build Tasks (complete every item)

- [] Build /assistant chat UI with message history
- [] Implement POST /api/ai/chat with Gemini function calling
- [] Tool: findNearbyIssues(lat, lng, radius_km, status?)
- [] Tool: getIssueById(issue_id)
- [] Tool: searchIssues(query, ward_id?)
- [] Tool: getHotspots(ward_id)
- [] Tool: getMyReports(user_id from auth)
- [] Tool: getDepartmentInfo(department_id)
- [] Tool: explainStatus(status enum)
- [] System prompt Appendix V: Never invent data; always call tools
- [] Bilingual EN/HI response based on user message language
- [] Rate limit 20 chat requests/min (Section 26)

Deliverables

- Chat answers 'issues near me' with real Firestore data
- Chat explains issue status for a given ID

Definition of Done

- [] Ask about fake issue ID -> tool returns NOT_FOUND not hallucination
- [] Nearby search returns actual map issues within radius

Verification Before Next Phase

- [] Demo script step 2:10 - chat query during judge demo (Appendix I)

PHASE 12: ISSUE THREADS & OPEN311 INTEROPERABILITY

Goal: Thread clustering, Gemini thread summaries, Open311 export adapter.

Suggested timing: Day 6 H3-6 | Section 36 Day 6

Depends on: Phase 5, 6 complete

Master Plan Coverage (sections + appendices)

- Section 11 - Open311 GeoReport v2 interoperability
- Section 23 - Thread clustering (CivicThreads pattern)
- Section 27 - GET /threads, POST /open311/export/:id, GET /departments
- Appendix M - Open311 service code mapping table
- Section 22 Route /threads/:id

Build Tasks (complete every item)

- [] Collection threads with issueIds[], summary, centroid, category
- [] Cluster issues by embedding similarity + proximity into threads
- [] Gemini auto-summary on thread update (CivicThreads pattern Section 3.2)
- [] Build Thread Detail /threads/:id with linked issues and AI overview
- [] Implement Open311 export POST /open311/export/:id (Section 11)
- [] Map internal categories to Appendix M service codes 001-050
- [] GET /departments returns Open311-compatible service catalog
- [] Document Open311 compatibility in docs/miscellaneous/open311.md
- [] Optional: GET /api/threads on map as heatmap overlay

Deliverables

- Related issues grouped into viewable threads
- Open311 JSON export for any issue ID
- Department service catalog API

Definition of Done

- [] Export JSON validates against Open311 GeoReport v2 field names
- [] Thread summary updates when new related issue added

Verification Before Next Phase

- [] Export one issue -> inspect service_code matches Appendix M

PHASE 13: PERFORMANCE, CACHING, SECURITY & RESILIENCE

Goal: L1-L5 caching, rate limits, waiting room, error codes, production hardening.

Suggested timing: Day 6 H6-9 | Section 36 Day 6

Depends on: Phases 2-12 functional

Master Plan Coverage (sections + appendices)

- Section 4.3-4.4 - Latency budget and optimizations
- Section 16 - All risks and mitigations implemented
- Section 25 - Performance & Caching L1-L5 full implementation
- Section 26 - Security 7 layers + waiting room
- Appendix W - All error codes and HTTP statuses
- Appendix G - Hardened security rules review
- Section 22 Route /waiting - WaitingRoom page

Build Tasks (complete every item)

- ☐ L1 RequestContext per-request geocode cache (Section 25)
- ☐ L2 In-memory Gemini cache by image SHA256 TTL 1hr max 200 entries
- ☐ L3 Redis optional for rate limits and analytics cache (Section 25)
- ☐ L4 analytics_daily pre-aggregation (verify Phase 8)
- ☐ L5 Static ward GeoJSON and taxonomy in /public
- ☐ express-rate-limit: 10 reports/day, 30 upvotes/hr, 20 chat/min (Section 26)
- ☐ Vision SafeSearch on all uploads (Section 26 Layer 4)
- ☐ Build /waiting page for 429/503 with retry countdown (LogiFlow pattern)
- ☐ Implement all Appendix W error codes with consistent JSON error body
- ☐ NEEDS_REVIEW 202, DUPLICATE_SUGGESTED 200, INVALID_MEDIA 400, etc.
- ☐ Parallel Promise.all upload+geocode (Section 4.4)
- ☐ P95 latency test: 10 concurrent reports < 5s (Section 33)
- ☐ Mitigate Section 16 risks: JSON mode uses flash not flash-image, Maps key restricted
- ☐ Health check GET /api/health with firestore connected status

Deliverables

- Rate limiting active on production
- Waiting room UX for overload
- Latency within Section 4.3 budget

Definition of Done

- ☐ No API keys in client bundle verified
- ☐ 429 returns proper Retry-After header

Verification Before Next Phase

- ☐ Run 11th report in one day -> rate limited
- ☐ Gemini cache hit on identical image re-upload shows lower latency

PHASE 14: FRONTEND COMPLETE - ALL 18 ROUTES & PWA POLISH

Goal: Finish every route from Section 22, all Appendix E screens, mobile PWA, accessibility.

Suggested timing: Day 6 H9-12 | Section 36 Day 6

Depends on: Phases 3-13 complete

Master Plan Coverage (sections + appendices)

- Section 22 - All 18 routes complete with Zustand stores
- Appendix E - All 11 screens polished
- Section 20.4 - Honest empty states on every list view
- Section 4 - Mobile-first PWA, offline draft queue IndexedDB stretch

Official Hackathon Features Addressed This Phase

- All 8 features visible in UI somewhere

Build Tasks (complete every item)

- [] Verify all routes: /, /report, /map, /issues/:id, /my-reports, /dashboard, /leaderboard, /assistant, /admin, /admin/analytics, /threads/:id, /login, /profile, /terms, /privacy, /waiting
- [] useAuthStore, useIssueStore, useMapStore, useDashboardStore wired (Section 22)
- [] Mobile responsive QA on iPhone Safari and Android Chrome
- [] PWA manifest + service worker for install prompt stretch
- [] WCAG AA contrast on severity badges (Section 33)
- [] Alt text on all issue images
- [] Loading skeletons and error boundaries on every async view
- [] IndexedDB offline draft queue for report in progress stretch
- [] Legal pages /terms /privacy
- [] Consistent design system: colors, typography, severity chips
- [] 404 page for unknown routes

Deliverables

- Every Section 22 route navigable
- Mobile demo-ready UI
- No broken links or blank screens

Definition of Done

- [] Complete user journey without dead ends: landing -> report -> map -> detail -> dashboard
- [] Appendix E all 11 screens implemented

Verification Before Next Phase

- [] Click through all nav links on mobile viewport 375px width
- [] Browser MCP E2E smoke test on deployed URL

PHASE 15: DOCUMENTATION, DIAGRAMS & LOGIFLOW PARITY

Goal: Complete docs/ folder, 16 mermaid diagrams, api_contract, LogiFlow-grade documentation.

Suggested timing: Day 6 H4-8 (parallel with Phase 14) | Section 36 Day 6

Depends on: Architecture stable from Phases 1-13

Master Plan Coverage (sections + appendices)

- Section 17 - Reference links in docs/README.md
- Section 19 - LogiFlow parity targets (8000+ doc lines goal)
- Section 30 - Full repository structure documentation
- Section 32 - All 16 diagram specifications
- Appendix Q - Match LogiFlow documentation depth
- Appendix S - Mermaid 01-system-architecture source
- Appendix T - Update TODO.md with completed items
- docs/pipelines/ - intake, vision, geo, verification, insights.md

Build Tasks (complete every item)

- [] Write docs/architecture.md from Sections 21, 22, 23 (LogiFlow architecture.md style)
- [] Write docs/system-design.md from Sections 20, 25 (LogiFlow system-design.md style)
- [] Write docs/deployment.md from Section 31
- [] Write docs/miscellaneous/api_contract.md from Section 27 + all appendices C,K,W
- [] Create docs/diagrams/mermaid/ 01 through 16 per Section 32 list
- [] Run render-diagrams.sh -> docs/diagrams/png/ exports
- [] docs/diagrams/README.md index like LogiFlow
- [] docs/ppt-info/slides/ 01-15 from Appendix P
- [] docs/README.md central index like LogiFlow docs/README.md
- [] Update README.md with architecture PNG, user journey PNG, live demo links
- [] Embed diagram 04-agent-workflow and 01-system-architecture in README
- [] Target 8000+ lines documentation cumulative (Section 19 stretch)

Deliverables

- docs/ folder matches Section 30 structure
- 16 PNG diagrams exported
- api_contract.md complete with every endpoint

Definition of Done

- [] docs/README.md links all documents
- [] Every Section 32 diagram has .mmd source and .png export

Verification Before Next Phase

- [] Compare docs structure side-by-side with LogiFlow-Solution-Challenge-2026/docs/

PHASE 16: DEVOPS, CI/CD & CLOUD DEPLOYMENT

Goal: Publish AI Studio to Cloud Run, gcloud deploy script, GitHub Actions, env prod, health monitoring.

Suggested timing: Day 6 H8-10 + Day 7 H0-1 | Section 36 Day 6-7

Depends on: Phases 1-15 complete

Master Plan Coverage (sections + appendices)

- Section 8.2-8.5 - AI Studio build, publish, deployment checklist
- Section 9 - gcloud CLI, firebase deploy, github MCP
- Section 31 - Full deployment runbook all paths A/B/C
- Section 35 - Makefile deploy, health targets
- Appendix N - Production env vars set in Cloud Run
- Section 16 - Deployment URL downtime risk mitigation

Build Tasks (complete every item)

- ☐ PATH A: AI Studio Publish -> Get Started -> Publish App -> copy Cloud Run URL
- ☐ PATH B: Standard tier with project-6d6f652b-7066-4341-806 if Starter ineligible
- ☐ PATH C: scripts/deploy-cloud-run.sh gcloud run deploy civicpulse-api asia-south1
- ☐ Cloud Run profile: 1 CPU, 512Mi-1Gi, min-instances 0 or 1, max 3, timeout 60s (Section 31)
- ☐ Set all Appendix N env vars on Cloud Run service
- ☐ Restrict MAPS_API_KEY to production URL + localhost
- ☐ Create .github/workflows/deploy.yml on push to main
- ☐ Create .github/workflows/lint.yml
- ☐ make deploy and make health commands working
- ☐ UptimeRobot or manual cron ping GET /api/health every 5 min during evaluation
- ☐ Verify camera + GPS on HTTPS production URL (Section 8.5 checklist)
- ☐ Do NOT delete AI Studio app until evaluation ends (Section 16)
- ☐ Document production URL in README and docs/deployment.md

Deliverables

- Stable public HTTPS Cloud Run URL
- GitHub Actions CI green
- deployment.md with exact URLs and env table

Definition of Done

- ☐ Section 8.5 deployment checklist all boxes checked
- ☐ Production URL loads on mobile with camera permission

Verification Before Next Phase

- ☐ curl production /api/health -> status ok
- ☐ Submit real report on production URL end-to-end

PHASE 17: TESTING, QA & DEMO SEED DATA

Goal: Unit/integration/E2E tests, seed 25 demo issues, accessibility audit, load smoke test.

Suggested timing: Day 6 H10-12 + Day 7 H1-2 | Section 36 Day 6-7

Depends on: Phase 16 deployed to production

Master Plan Coverage (sections + appendices)

- Section 33 - Full testing & QA strategy
- Appendix R - Seed 25 demo issues specification exactly
- Appendix I - Rehearse demo on seeded data
- Section 13 - QA before submission

Build Tasks (complete every item)

- ☐ Unit tests: Zod schema, priority score formula, SLA deadline calc, geohash encode
- ☐ Integration test: POST /api/reports fixture image -> expect 201 + schema fields
- ☐ Agent tests: mock Gemini; verify low confidence -> review queue branch
- ☐ E2E via browser MCP: full report flow on production URL
- ☐ scripts/seed-demo-issues.ts: 25 issues per Appendix R breakdown
 - ☐ 5 potholes, 4 waste, 3 streetlights, 3 water leaks, 2 road_damage, 2 drainage, 2 resolved with proof, 2 community-verified
- ☐ All isDemo: true, wardId DEMO_WARD_001, within 2km cluster
- ☐ Load smoke: 10 concurrent POST /api/reports P95 < 5s
- ☐ Manual QA checklist Section 33: mobile Safari, Android Chrome, admin flow
- ☐ WCAG AA audit on severity colors
- ☐ Fix all P0 bugs before Phase 18

Deliverables

- 25 demo issues on production for judge exploration
- Test suite passing in CI
- QA sign-off checklist completed

Definition of Done

- ☐ Demo ward shows rich map cluster for judges
- ☐ No P0 bugs open

Verification Before Next Phase

- ☐ Run Appendix I demo script once on production with seed data
- ☐ make test passes locally and in CI

PHASE 18: SUBMISSION PACKAGE - GITHUB, GOOGLE DOC, BLOCKSEBLOCK

Goal: Prepare and submit all mandatory hackathon deliverables with presentation kit.

Suggested timing: Day 7 H1-3 | Section 36 Day 7 | DEADLINE 2:00 PM

Depends on: Phases 16-17 complete

Master Plan Coverage (sections + appendices)

- Section 14 - All mandatory submission deliverables
- Section 12 - Evaluation matrix addressed in Google Doc narrative
- Appendix J - Google Doc 10-section template
- Appendix P - 15-slide presentation kit
- Appendix A - Evaluation criteria explicitly mapped in doc
- Appendix F - Competitive matrix in doc
- Section 34 - Business impact metrics in doc
- BlockseBlock submission steps from hackathon PDF

Official Hackathon Features Addressed This Phase

- Document all 8 official features with screenshots in Google Doc

Build Tasks (complete every item)

- ☐ GitHub repo final polish: README with live URL, architecture diagram, team, stack
- ☐ Tag release v1.0.0-submission
- ☐ Write Google Doc per Appendix J all 10 sections with public link
- ☐ Section 1: Problem Statement Selected - Community Hero
- ☐ Section 2: Solution Overview 300 words
- ☐ Section 3: Key Features - all 8 with screenshots from production
- ☐ Section 4: Technologies Used
- ☐ Section 5: Google Technologies Utilized (AI Studio, Gemini, Cloud Run, Firestore, Auth, Storage, Maps)
- ☐ Section 6-7: Embed architecture + agent workflow PNGs
- ☐ Section 8: Live URL + GitHub URL
- ☐ Section 9: Team members and roles (Appendix U)
- ☐ Section 10: Future roadmap ADK, Open311, multilingual voice
- ☐ Build 15 slides from Appendix P in Google Slides or PPT
- ☐ BlockseBlock: Dashboard -> Create Project -> select Community Hero problem
- ☐ Enter Deployed Application Link (Cloud Run URL)
- ☐ Enter GitHub Repository Link
- ☐ Enter Google Doc Link (anyone with link can view)
- ☐ Toggle both notes -> Final Submit (irreversible - verify first)
- ☐ Verify all links work in incognito browser

Deliverables

- Three submission links live and accessible
- BlockseBlock shows submitted status
- Google Doc version history available for jury

Definition of Done

- ☐ Section 14 checklist fully checked
- ☐ All three mandatory links tested incognito
- ☐ Submitted before June 29 2026 2:00 PM

Verification Before Next Phase

- ☐ Open all 3 links on phone without login - all load
- ☐ Google Doc covers every evaluation criterion from Appendix A

PHASE 19: FINAL DEMO, LAUNCH & PROJECT CLOSURE

Goal: Rehearse judge demo, mentor-ready walkthrough, monitor production, close project with retrospective.

Suggested timing: Day 7 H0-2 rehearsal + post-submit monitoring | Section 36 + Section 18

Depends on: Phase 18 submitted

Master Plan Coverage (sections + appendices)

- Section 18 - Conclusion and winning formula delivered
- Appendix I - 3-minute judge demo script (full rehearsal twice)
- Section 36 Day 7 - Final buffer and deadline
- Appendix T - All TODO items checked
- Hackathon mentor session insights applied (June 24 retrospective)

Build Tasks (complete every item)

- [] Rehearse Appendix I demo script twice timed (target under 3 minutes)
- [] 0:00 landing stats -> 0:20 report photo -> 0:45 AI fill -> 1:10 map pin
- [] 1:30 second device upvote -> 1:50 dashboard -> 2:10 AI chat -> 2:30 admin resolve -> 2:50

GitHub+Doc+stack

- [] Prepare QR code slide pointing to production URL for jury
- [] Monitor Cloud Run logs during evaluation period
- [] Keep AI Studio app published - do not delete
- [] Respond to any jury requests for additional evidence
- [] Team retrospective: what shipped vs master plan
- [] Mark all Phase 0-19 checkboxes complete in this PDF
- [] Optional: post-hackathon roadmap - ADK full deploy, Capacitor APK, BigQuery, Vertex fine-tuning
- [] Celebrate - CIVICPULSE AI shipped

Deliverables

- Flawless 3-minute demo capability
- Production stable through evaluation window
- All phase checkboxes complete

Definition of Done

- [] Demo rehearsed twice under 3 minutes
- [] Appendix T TODO.md all items checked
- [] Section 18 conclusion vision achieved: shipped not slideware

Verification Before Next Phase

- [] Record demo video as backup for presentation
- [] Final read of Section 18 winning formula - did we deliver agentic depth + Google stack + impact?

ROLLING MASTER CHECKLIST (APPENDIX T - ALL PHASES)

Copy to repo TODO.md. Check each item as the corresponding phase completes.

PHASE 0 COMPLETE: FOUNDATION, REQUIREMENTS & QUALITY BAR

- ☐ P0 Done: Every team member read Master Plan Sections 1-3 and 12
- ☐ P0 Done: All 8 hackathon features mapped to phases 2-12
- ☐ P0 Done: Evaluation criteria understood and design decisions will target Agentic Depth 20%

PHASE 1 COMPLETE: PROJECT SCAFFOLD & GOOGLE CLOUD FOUNDATION

- ☐ P1 Done: AI Studio Build mode loads without errors
- ☐ P1 Done: Google Sign-In works in dev container
- ☐ P1 Done: GitHub repo exists and is pushed
- ☐ P1 Done: docs/ and scripts/ folders match Section 30 structure

PHASE 2 COMPLETE: REPORT INTAKE + GEMINI VISION AI PIPELINE

- ☐ P2 Done: End-to-end: photo -> AI analysis -> editable form -> Firestore write
- ☐ P2 Done: P95 classify latency < 4s on test image
- ☐ P2 Done: Invalid/blank images rejected with InvalidMediaCard (Section 20.4)

PHASE 3 COMPLETE: GEO-LOCATION, MAPS & ISSUE DISCOVERY UI

- ☐ P3 Done: Map loads on mobile Safari/Chrome over HTTPS
- ☐ P3 Done: Filters correctly hide/show issue categories
- ☐ P3 Done: Empty state: 'Be the first reporter' when no issues (Section 20.4)

PHASE 4 COMPLETE: FIRESTORE DATA LAYER, REALTIME SYNC & ISSUE DETAIL

- ☐ P4 Done: Status transitions write events subcollection entries
- ☐ P4 Done: Unauthorized users cannot update others' issues
- ☐ P4 Done: My Reports shows only current user's issues

PHASE 5 COMPLETE: COMMUNITY VERIFICATION & DUPLICATE DETECTION

- ☐ P5 Done: Second account upvote increments count in realtime
- ☐ P5 Done: Duplicate report within 50m suggests merge rather than duplicate pin spam
- ☐ P5 Done: Rate limit returns 429 with Retry-After (Appendix W)

PHASE 6 COMPLETE: AGENTIC AI ORCHESTRATION (6 AGENTS)

- ☐ P6 Done: Low confidence report goes to review queue not public map
- ☐ P6 Done: SLA deadline visible on issue detail
- ☐ P6 Done: Agentic workflow diagram ready for docs (content for Phase 15)

PHASE 7 COMPLETE: ADMIN WORKFLOW, RESOLUTION & SLA ENFORCEMENT

- ☐ P7 Done: Non-admin gets 403 on status update endpoint
- ☐ P7 Done: Resolved issue shows proof image on detail page
- ☐ P7 Done: Reopen flow returns issue to Submitted status

PHASE 8 COMPLETE: IMPACT DASHBOARDS & ANALYTICS

- ☐ P8 Done: KPI numbers match manual Firestore count verification
- ☐ P8 Done: Dashboard performs acceptably with 25+ seed issues

PHASE 9 COMPLETE: PREDICTIVE INSIGHTS & HOTSPOT INTELLIGENCE

- ☐ P9 Done: Cluster of 5+ issues triggers hotspot alert
- ☐ P9 Done: Gemini narrative mentions actual category counts from data not hallucinated

PHASE 10 COMPLETE: GAMIFICATION & CIVIC ENGAGEMENT

[] P10 Done: Gamification cannot be gamed without real reports (rate limits)

[] P10 Done: Leaderboard hidden unless user opts in

PHASE 11 COMPLETE: CITIZEN AI ASSISTANT (FUNCTION CALLING CHAT)

[] P11 Done: Ask about fake issue ID -> tool returns NOT_FOUND not hallucination

[] P11 Done: Nearby search returns actual map issues within radius

PHASE 12 COMPLETE: ISSUE THREADS & OPEN311 INTEROPERABILITY

[] P12 Done: Export JSON validates against Open311 GeoReport v2 field names

[] P12 Done: Thread summary updates when new related issue added

PHASE 13 COMPLETE: PERFORMANCE, CACHING, SECURITY & RESILIENCE

[] P13 Done: No API keys in client bundle verified

[] P13 Done: 429 returns proper Retry-After header

PHASE 14 COMPLETE: FRONTEND COMPLETE - ALL 18 ROUTES & PWA POLISH

[] P14 Done: Complete user journey without dead ends: landing -> report -> map -> detail -> dashboard

[] P14 Done: Appendix E all 11 screens implemented

PHASE 15 COMPLETE: DOCUMENTATION, DIAGRAMS & LOGIFLOW PARITY

[] P15 Done: docs/README.md links all documents

[] P15 Done: Every Section 32 diagram has .mmd source and .png export

PHASE 16 COMPLETE: DEVOPS, CI/CD & CLOUD DEPLOYMENT

[] P16 Done: Section 8.5 deployment checklist all boxes checked

[] P16 Done: Production URL loads on mobile with camera permission

PHASE 17 COMPLETE: TESTING, QA & DEMO SEED DATA

[] P17 Done: Demo ward shows rich map cluster for judges

[] P17 Done: No P0 bugs open

PHASE 18 COMPLETE: SUBMISSION PACKAGE - GITHUB, GOOGLE DOC, BLOCKSEBLOCK

[] P18 Done: Section 14 checklist fully checked

[] P18 Done: All three mandatory links tested incognito

[] P18 Done: Submitted before June 29 2026 2:00 PM

PHASE 19 COMPLETE: FINAL DEMO, LAUNCH & PROJECT CLOSURE

[] P19 Done: Demo rehearsed twice under 3 minutes

[] P19 Done: Appendix T TODO.md all items checked

[] P19 Done: Section 18 conclusion vision achieved: shipped not slideware

APPENDIX: MASTER PLAN SECTION QUICK REFERENCE

Sections 1-36 and Appendices A-W are fully distributed across Phases 0-19.

If unsure where a topic lives, consult the Coverage Index at the start of this PDF.

Source document: Community-Hero-Master-Plan.pdf (~61 pages, ~36 sections, 23 appendices).

This phase plan adds: timelines, checkboxes, dependencies, verification gates, and 100% coverage mapping.

PROJECT END: Phase 19 complete + BlockseBlock submitted = Community Hero project delivered.